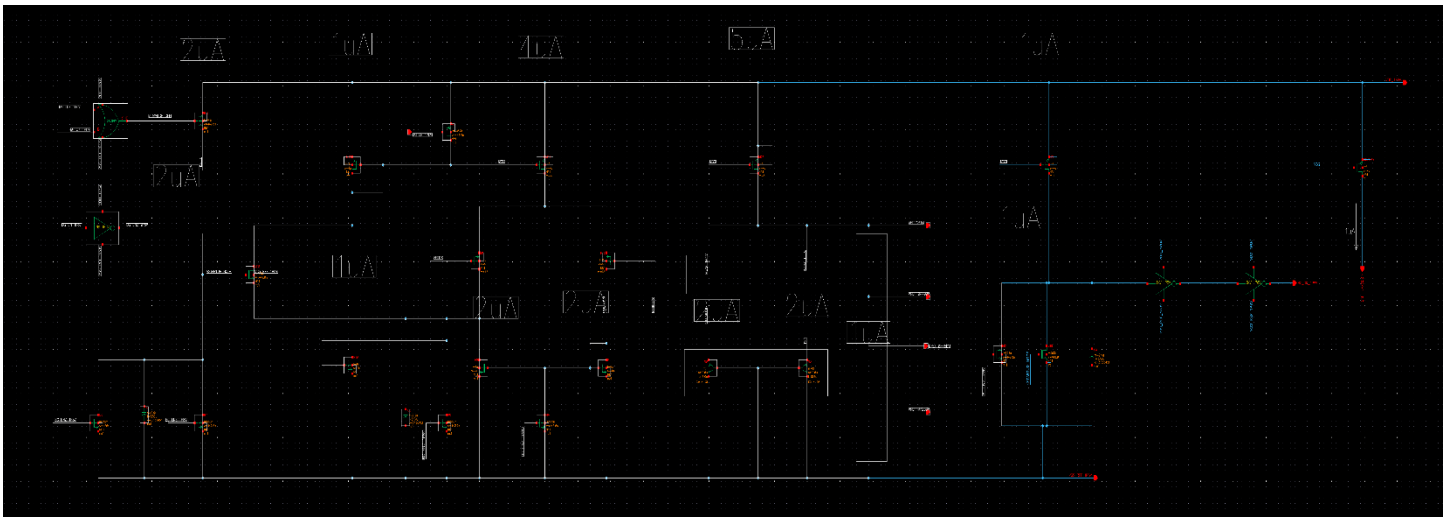
Bandgap Reference (BGR)

This report presents the design and layout of a Bandgap Reference (BGR) circuit. It includes the schematic & explanation and layout guidelines information like device placement, bias circuitry, and routing strategy. Additional focus is on device matching, layout symmetry, parasitic reduction, and proper routing techniques to improve reference voltage stability, minimize mismatch, and enhance noise immunity.

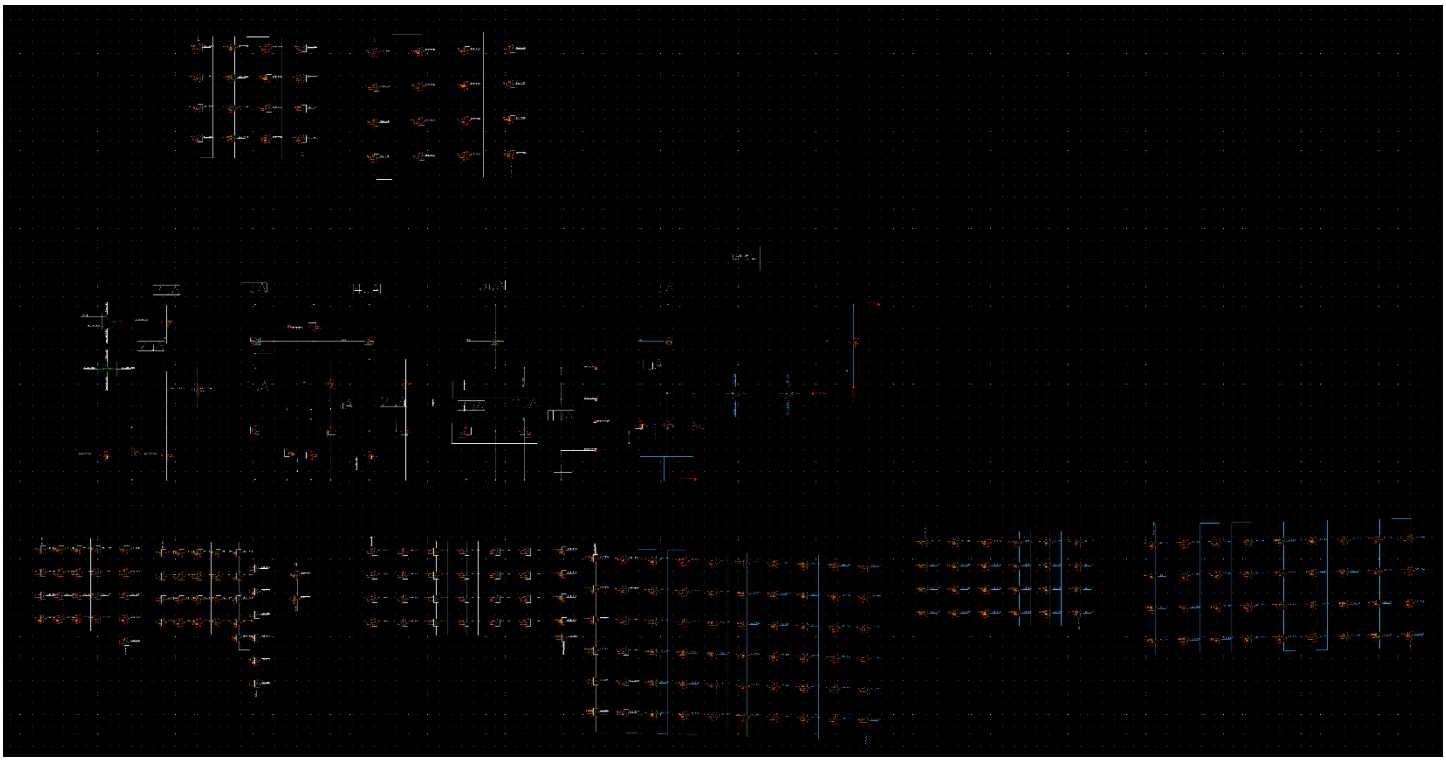
What is BGR?

A Bandgap Reference (BGR) is an analog circuit that generates a nearly constant reference voltage, independent of changes in temperature, supply voltage, and manufacturing process variations.

Schematic of BGR



Overall schematic with Resistor connection



The main block of a Bandgap Reference (BGR) circuit generally consists:

1. **Startup Circuit**

- Purpose: Ensures the BGR starts operating when the power supply is switched on.
- Working: Some BGR circuits can remain in an unwanted zero-current state. The startup circuit injects a small current to move the circuit into its correct operating condition.
- Layout consideration: Startup transistors should be placed and routed carefully to avoid unwanted interference with sensitive analog nodes.

2. **Current Mirror Circuit**

- Purpose: Copies a reference current into different branches of the BGR.
- Working: When matched MOSFETs operate under suitable conditions, their drain currents can be approximately equal. The current is determined by the transistor dimensions, bias conditions, and other circuit parameters.
- Layout consideration: Matched transistors should have similar surroundings and may use common-centroid or interdigitated placement to reduce mismatch.

3. **BJT / Parasitic BJT Block**

- Purpose: Generates the base-emitter voltage (V_{BE}).
- Working: The base-emitter voltage of a BJT generally decreases as temperature increases. Therefore, it provides a CTAT voltage (Complementary to Absolute Temperature).
- Layout consideration: If multiple BJTs are used, their relative emitter areas and operating current densities must be controlled to generate an accurate voltage difference.

4. **PTAT Generation Block**

- Purpose: Generates a voltage proportional to absolute temperature.
- Working: Two BJTs operating at different current densities produce a difference in base-emitter voltage:

$$\Delta V_{BE} = \frac{kT}{q}\ln(N)$$

Where:

- k = Boltzmann constant
- T = Absolute temperature in kelvin
- q = Electron charge
- N = Ratio of the BJTs' current densities, under the usual ideal assumptions

As temperature increases, V_{BE} increases.

- Layout consideration: Maintain the intended BJT area ratio and use symmetric routing where practical.

5. CTAT Generation Block

- Purpose: Provides a voltage that decreases with increasing temperature.
- Working: The BJT base-emitter voltage \(V_{BE}\) generally decreases as temperature rises.
- Layout consideration: Ensure that the BJT bias conditions remain correct over the intended operating range.

6. Op-Amp / Error Amplifier

- Purpose: Maintains the required voltage relationship between two circuit nodes.
- Working: The amplifier senses the voltage difference between its inputs and adjusts its output to reduce that difference through feedback.
- Layout consideration: Use suitable matching techniques for the input differential pair, maintain symmetry, and minimize parasitic effects on sensitive nodes.

Note: Not every BGR design requires an op-amp; some use other biasing and feedback arrangements.

7. Resistor Network

- Purpose: Sets current levels and scales the PTAT voltage contribution.
- Working: Resistors convert current into voltage and establish the ratios needed to combine the PTAT and CTAT components.
- Layout consideration: Resistor matching and accurate resistance ratios are important for reference-voltage accuracy.

8. Reference Voltage Output Block

- Purpose: Produces the final, temperature-compensated reference voltage.
- Working: The circuit combines the CTAT voltage with a scaled PTAT voltage:

$$V_{REF} = V_{BE} + K\Delta V_{BE}$$

Here, K is a design-dependent scaling factor.

When the two temperature coefficients are properly balanced, the resulting reference voltage varies relatively little with temperature.

Layout Techniques Used

1. **Matching Techniques**

    - Common-centroid placement
    - Interdigitation
    - Resistor matching
    - Symmetrical device placement
    - Dummy device placement
    - BJT matching

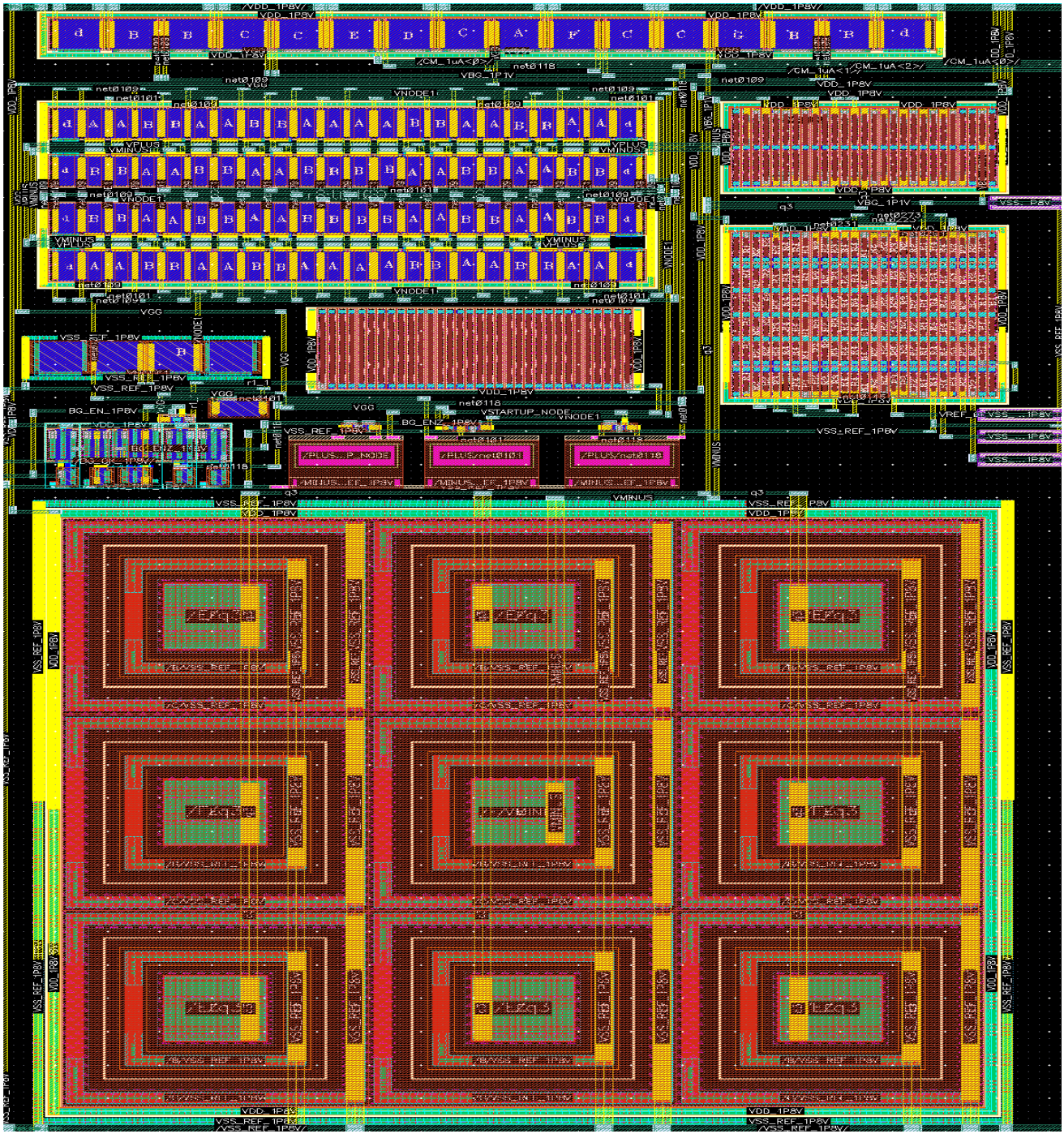
2. **Noise Reduction Techniques**

    - Guard ring implementation
    - Shielding of sensitive nets

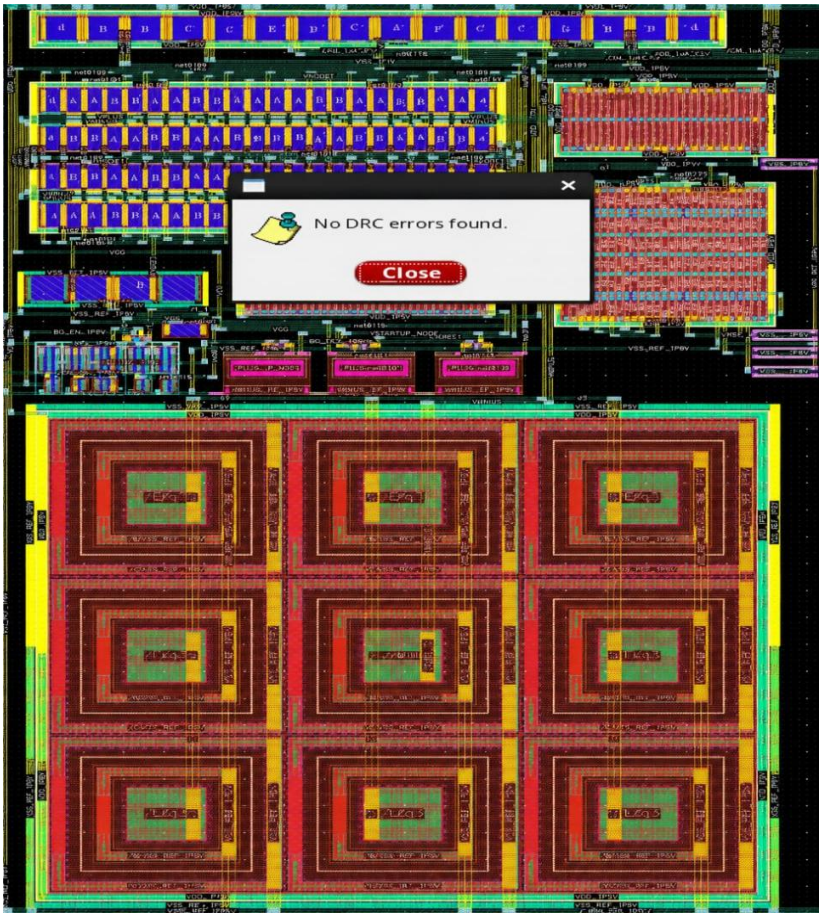
3. **Routing Optimization**

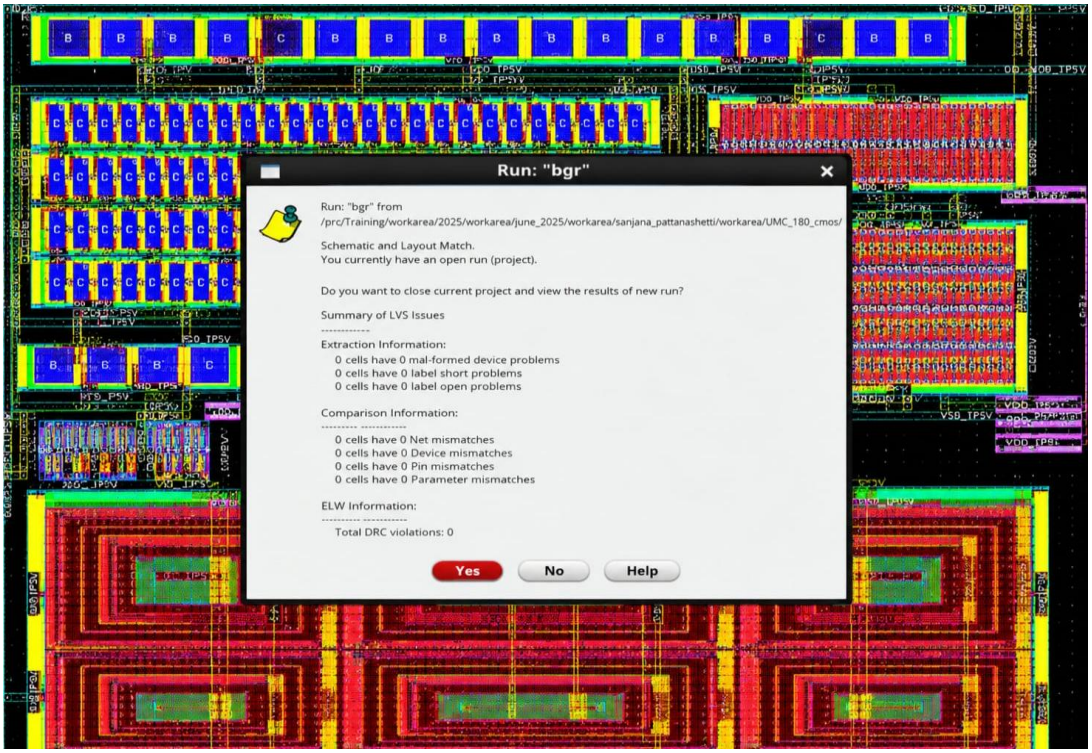
    - Short and symmetrical routing
    - Power-routing optimization

Layout of BGR



DRC and LVS Verification





Conclusion

The Bandgap Reference (BGR) layout was successfully designed using Cadence Virtuoso with proper device matching, symmetry, and routing techniques. Common-centroid placement, interdigitation, shielding and guard rings were implemented to improve matching accuracy and reduce noise.

The layout passed DRC verification with zero violations and LVS verification with no mismatches, ensuring layout correctness and consistency with the schematic.